So far:

Techniques for integration → ① use formulas directly

→ ② Substitution

Next:

→ ③ Integration by Parts.

Product rule:

$$[f(x)\cdot g(x)]' = f'(x)\cdot g(x) + g'(x)\cdot f(x)$$

$$\Rightarrow \int [f(x)\cdot g(x)]' = \int f'(x)\cdot g(x) + \int g'(x)\cdot f(x)$$

$$\Rightarrow f(x)\cdot g(x) = \int f'(x)\cdot g(x) + \int g'(x)\cdot f(x)$$

$$\Rightarrow \int f'(x)\cdot g(x) = f(x)\cdot g(x) - \int g'(x)\cdot f(x)$$

7.2: Integration by parts

Formula:

$$\int u\cdot dv = u\cdot v - \int v\cdot du$$

Example

①

$$I = \int x \cdot e^x \, dx$$

$$\begin{cases} u = x \\ dv = e^x \, dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \int e^x \, dx = e^x \end{cases}$$

$$I = \int u \cdot dv = u \cdot v - \int v \cdot du$$

$$\Rightarrow \quad I = x \cdot e^x - \int e^x \, dx$$

$$= x \cdot e^x - e^x + C$$

②

$$I = \int x \cdot \sin x \cdot dx$$

$$\begin{cases} u = x \\ dv = \sin x \cdot dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \int \sin x = -\cos x \end{cases}$$

$$\Rightarrow$$

$$I = u \cdot v - \int v \cdot du$$

$$= x \cdot (-\cos x) - \int -\cos x \cdot dx$$

$$= -x\cos x + \sin x + C$$

③ $I = \int x \cdot \ln x \, dx$

$$\begin{cases} u = x \\ dv = \ln x \, dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \int \ln x \, dx \end{cases}$$

OK $I = \int (\ln x) \cdot x \cdot dx$

$$\begin{cases} u = \ln x \\ dv = x \cdot dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{x} dx \\ v = \frac{x^2}{2} \end{cases}$$

$$\Rightarrow I = u \cdot v - \int v \cdot du$$

$$= (\ln x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= (\ln x) \cdot \frac{x^2}{2} - \frac{1}{2} \int x \, dx$$

$$= (\ln x) \cdot \frac{x^2}{2} - \frac{1}{4} x^2 + C$$

④ $I = \int 1 \cdot \ln x \cdot dx$

$$\begin{cases} u = \ln x \\ dv = dx \end{cases} \qquad \begin{cases} du = \frac{1}{x} dx \\ v = x \end{cases}$$

$$\Rightarrow I = (\ln x) \cdot x - \int x \cdot \frac{1}{x} dx$$

$$I = x \cdot \ln x - x + C$$

Practica :

$$\int x^2 \cdot e^x \cdot dx$$

$u =$ $\qquad$ $du =$

$dv =$ $\qquad$ $v =$